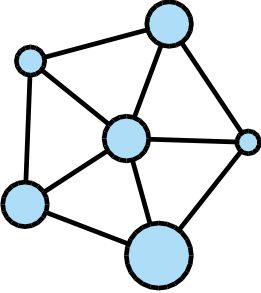
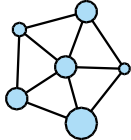
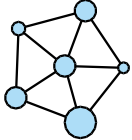


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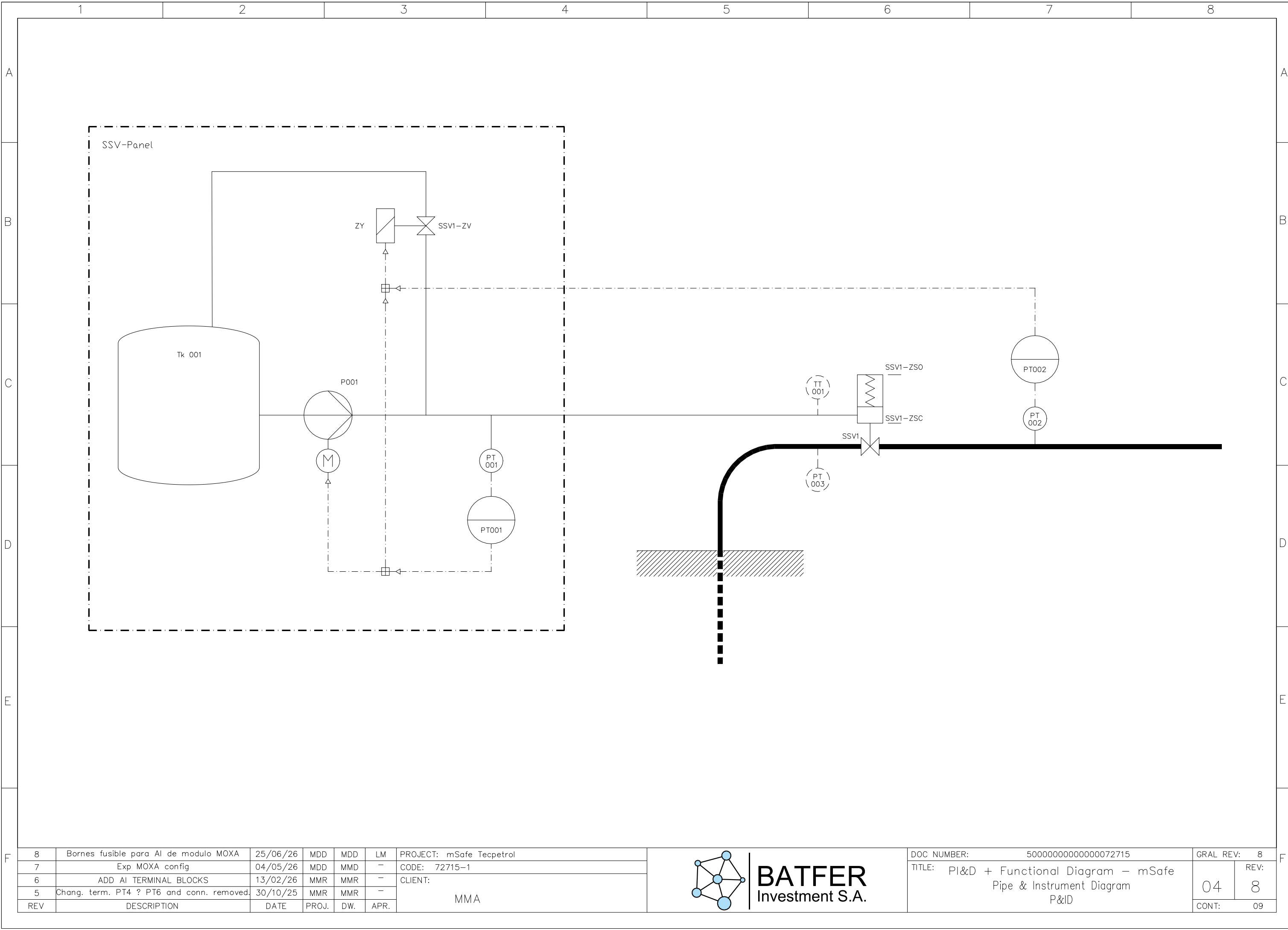
NOTE: cables not indicated will be black 0.75mm²/18 AWG

1	2	3	4	5	6	7	8																																				
A								A																																			
B	 BATFER Investment S.A.							B																																			
C								C																																			
D	Client: - MMA Project: - mSafe Tecpetrol Description Document: - 72715-1 Electrical Diagram mSafe Document Number: - 72715 ZPL Revision: - 8					APROBADO PARA PRODUCCION APROBADO: – FECHA: 25/06/2026  BATFER Investment S.A.			D																																		
E	Installation Place -							E																																			
F	<table><tr><td>8</td><td>Bornes fusible para AI de modulo MOXA</td><td>25/06/26</td><td>MDD</td><td>MDD</td><td>LM</td><td>PROJECT: mSafe Tecpetrol</td></tr><tr><td>7</td><td>Exp MOXA config</td><td>04/05/26</td><td>MDD</td><td>MMD</td><td>–</td><td>CODE: 72715–1</td></tr><tr><td>6</td><td>ADD AI TERMINAL BLOCKS</td><td>13/02/26</td><td>MMR</td><td>MMR</td><td>–</td><td>CLIENT:</td></tr><tr><td>5</td><td>Chang. term. PT4 ? PT6 and conn. removed.</td><td>30/10/25</td><td>MMR</td><td>MMR</td><td>–</td><td>MMA</td></tr><tr><td>REV</td><td>DESCRIPTION</td><td>DATE</td><td>PROJ.</td><td>DW.</td><td>APR.</td><td></td></tr></table>					8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol	7	Exp MOXA config	04/05/26	MDD	MMD	–	CODE: 72715–1	6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	–	CLIENT:	5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	–	MMA	REV	DESCRIPTION	DATE	PROJ.	DW.	APR.		 BATFER Investment S.A.	DOC NUMBER: 50000000000000072715 TITLE: PI&D + Functional Diagram – mSafe COVER – GRAL REV: 8 REV: 8 CONT: 01	F
8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol																																					
7	Exp MOXA config	04/05/26	MDD	MMD	–	CODE: 72715–1																																					
6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	–	CLIENT:																																					
5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	–	MMA																																					
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.																																						

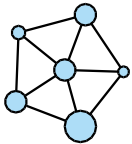
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1		2		3		4		5		6		7		8														
A	Pág.	Page Title		Revision								Additional Comments	Pág.	Page Title		Revision								Additional Comments				
				2	3	4	5	6	7	8	2					3	4	5	6	7	8							
		00	COVER				X	X	X	X					45	FREE												
01	INDEX				X	X	X	X			46	P&ID																
02	FREE										47	FREE																
03	FREE										48	FREE																
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05	FREE										50	FREE																
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07	FREE										52	FREE																
08	FREE										53	FREE																
09	Material's List										54	FREE																
10	FREE										55	FREE																
11	Power Generation				X						56	FREE																
12	Battery Chargers and Power Circuits			X	X						57	FREE																
13	Control Voltage – Generation and Distribution	X		X	X		X				58	FREE																
14	Control Unit – Power Supply										59	FREE																
15	Intrinsic Safety Barrier – Power Supply										60	FREE																
16	Common Distribution for Signals	X		X							61	Digital Outputs –Main Board			X													
17	FREE										62	FREE																
18	FREE										63	FREE																
19	FREE										64	FREE																
20	FREE										65	FREE																
21	Electronic Board (21PCB01) – Main Board			X	X						66	FREE																
22	FREE										67	FREE																
23	FREE										68	FREE																
24	FREE										69	FREE																
25	FREE										70	FREE																
26	FREE										71	FREE																
27	FREE										72	FREE																
28	FREE										73	FREE																
29	FREE										74	FREE																
30	FREE										75	FREE																
31	Inner Analog Inputs – Main Board			X	X						76	FREE																
32	External Analog Inputs – Main Board			X	X						77	FREE																
33	External Analog Inputs – Analog Input Module	X			X			X			78	FREE																
34	FREE										79	FREE																
35	FREE										80	FREE																
36	FREE										81	Communications		X			X											
37	FREE										82	FREE																
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40	FREE										85	FREE																
41	FREE										86	FREE																
42	FREE										87	FREE																
43	External Digital Inputs – Main Board				X						88	FREE																
44	FREE										89	FREE																
F	8	Bornes fusible para AI de modulo MOXA		25/06/26		MDD	MDD	LM	PROJECT: mSafe Tecpetrol										DOC NUMBER: 50000000000000072715								GRAL REV: 8	
	7	Exp MOXA config		04/05/26		MDD	MMD	–	CODE: 72715–1										TITLE: PI&D + Functional Diagram – mSafe								REV: 8	
	6	ADD AI TERMINAL BLOCKS		13/02/26		MMR	MMR	–	CLIENT: MMA										INDEX								01	
	5	Chang. term. PT4 ? PT6 and conn. removed.		30/10/25		MMR	MMR	–											–								CONT: 04	
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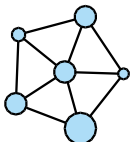
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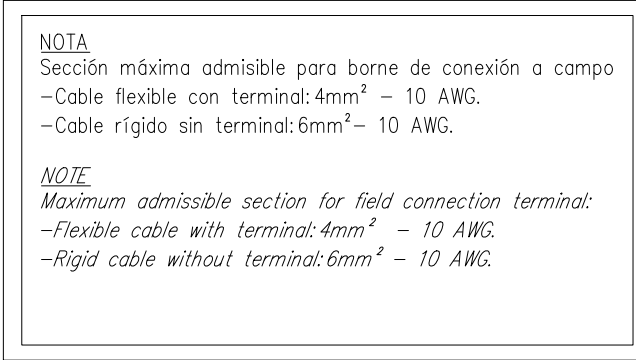


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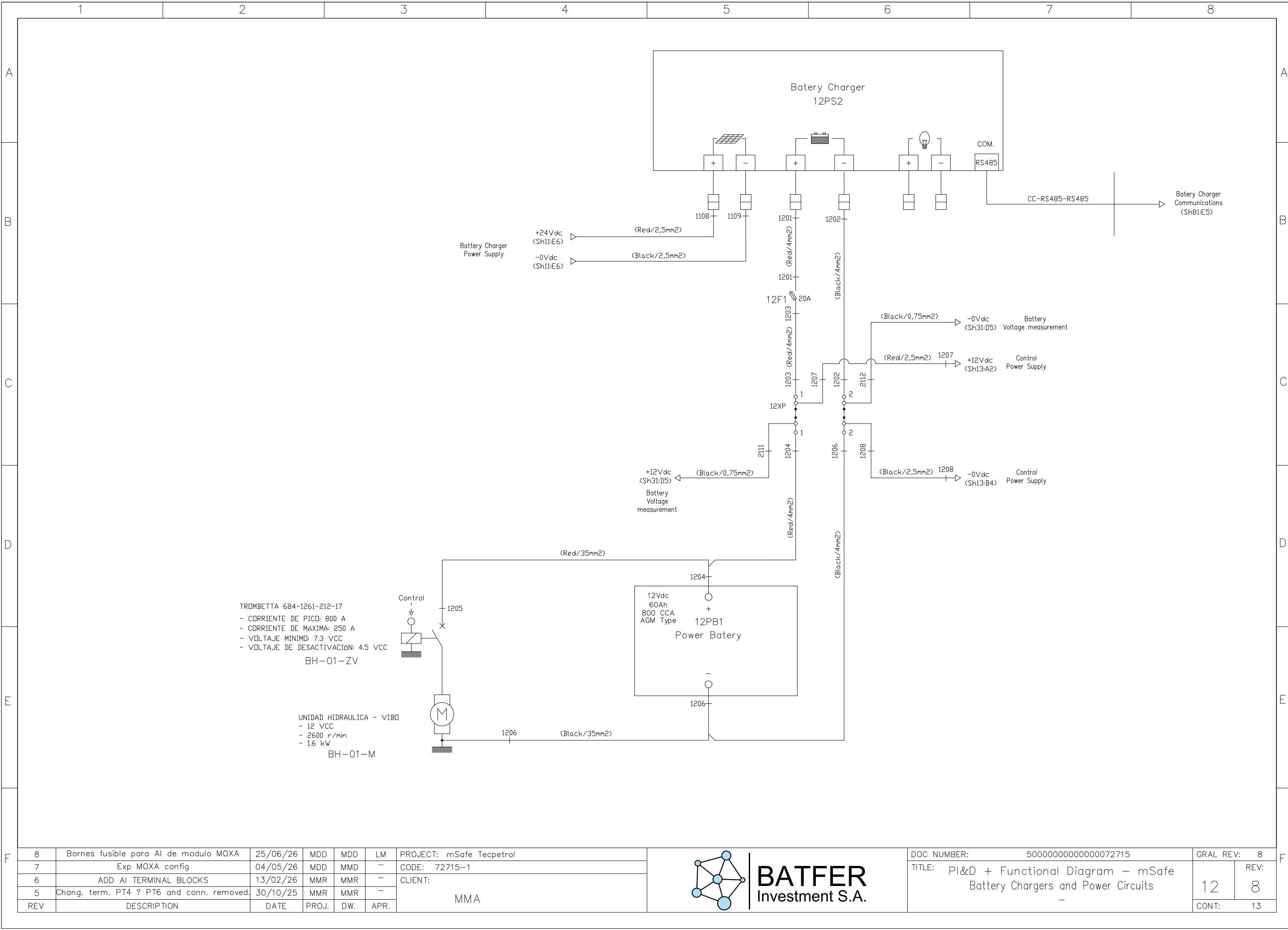
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TITLE:	PI&D + Functional Diagram - mSafe Pipe & Instrument Diagram P&ID

GRAL REV:	8
REV:	8
CONT:	09

1		2		3		4		5		6		7		8			
TAG		DESCRIPTION						BRAND				MODEL / CODE					
A	11XP	Power Supply Terminals / Bornes de alimentación						PHOENIX CONTACT				PT 4					
	11PS1	Power Source 110–240 Vca to 12 Vdc / Fuente de alimentación 110–240 Vac a 12 Vcc						MEAN WELL				NDR–120–12					
	11Q1	AC Control Circuit Breaker / Interruptor térmico de control AC						ABB				SH 202 – C10					
	12PS2	Battery Charger 12 Vdc / Cargador de baterías 12 Vcc						EPEVER				Tracer 5210BP					
	12XP	12Vdc Power Distribution terminal blocks / Bornes de distribución alimentación 12Vcc						PHOENIX CONTACT				PT 4					
	BH–01–M	Hidraulic Pump Motor / Motor de bomba hidráulica						BREVINI				--					
	BH–01–ZV	Hidraulic Pump Motor Starter / Arrancador motor de bomba hidráulica						BREVINI				--					
	12PB1	Power Battery / Batería de potencia						--				--					
B	12F1	Power Battery Charger Fuse / Base de fusible para cargador de baterías						ABB				E 91/32					
	13XC1	12Vdc Control Voltage Distribution / Distribución de voltaje de control 12Vcc						PHOENIX CONTACT				PT 4					
	13PS2	12/24Vdc Converter / Conversor 12/24Vcc						MEAN WELL				DDR–120A–24					
	13F2	12Vdc Control Voltage Fuse Base / Base de fusible para voltage de control 12Vcc						ABB				E 91/32					
	21PCB01	Principal Control Electronic board / Placa Electrónica de Control Principal						MMA				L03–60590–01					
	31XAI	Inner Analog Inputs Terminals / Bornes de entradas analógicas internas						PHOENIX CONTACT				PTT 2,5–2MT					
	PT–001	Hidraulic Pump Pressure Transmitter / Transmisor de presión de la bomba hidráulica						GEFRAN				KS					
	32XAI	Analog Inputs Terminals / Bornes de entradas analógicas						PHOENIX CONTACT				PTT 2,5–2MT					
	33XAI	Analog Inputs Terminals / Bornes de entradas analógicas						PHOENIX CONTACT				PTT 2,5–2MT					
	33MX01	Analog Inputs Module / Modulo de entradas analógicas						MOXA				ioLogik R1240–T					
C	43XDI	External Comand 24Vdc Terminals / Bornes de comando externo 24 Vcc						PHOENIX CONTACT				PTT 2,5–2MT					
	43DIB1	Digital Input signals – Intrinsic safety barrier / Entrada Digital – Barrera Seg. Intrínseca						CHENZHU				GS8512–EX.22					
	43XDIB	Intrinsic Safety Digital Inputs Terminals / Bornes de entradas digitales Seg. Intrínseca						PHOENIX CONTACT				PTT 2,5–2 MT BU					
	43KR	Terminal Relay 24Vdc / Borne relé 24Vcc						PHOENIX CONTACT				RIF–0–RPT–24DC/21					
	61KR1..3	Terminal Relay / Borne relé						PHOENIX CONTACT				RIF–0–RPT–12DC/21					
	XM1	Commands Terminals / Bornes de comandos						PHOENIX CONTACT				PTT 2,5–2MT					
	ZY	Solenoid Valve / Válvula solenoide						BREVINI				CRP04 NA 18W					
	81XCM	Communication Terminal / Borne de comunicación						PHOENIX CONTACT				PTT 2,5–2MT					
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8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol		BATFER Investment S.A.				DOC NUMBER: 5000000000000072715				GRAL REV: 8	
7	Exp MOXA config	04/05/26	MDD	MMD	–	CODE: 72715–1						TITLE: PI&D + Functional Diagram – mSafe				REV:	
6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	–	CLIENT:						Material's List				098	
5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	–	MMA						–				CONT: 11	
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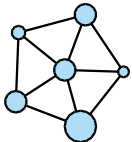


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6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	—	CLIENT:					
5	Chang. term. PT4 ? PT6 and conn. removed.	30/10/25	MMR	MMR	—	MMA					
REV	DESCRIPTION	DATE	PROJ.	DW.	APR.						



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8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol
7	Exp MOXA config	04/05/26	MDD	MMD	-	CODE: 72715-1
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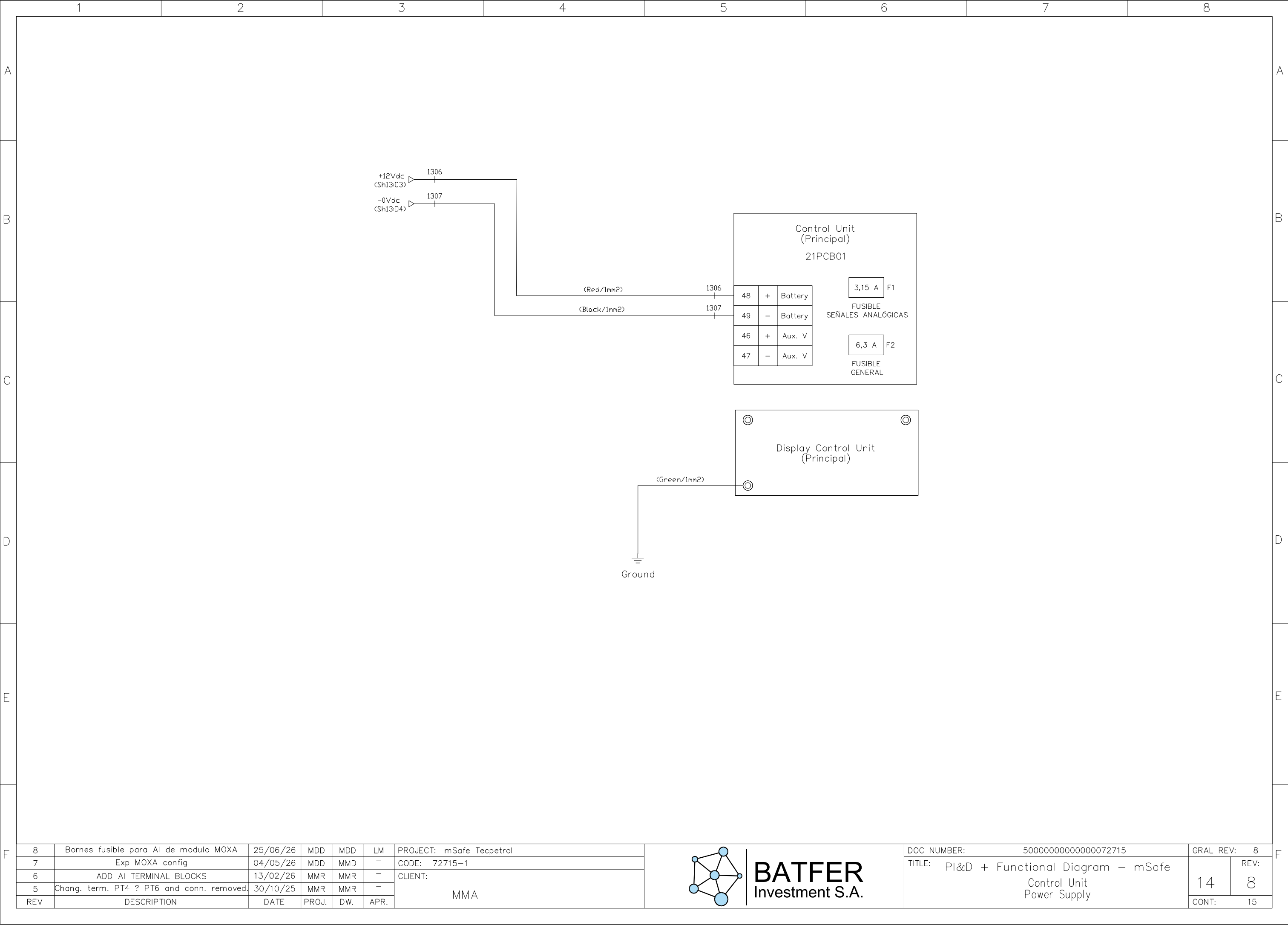
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REV:	8
CONT:	13

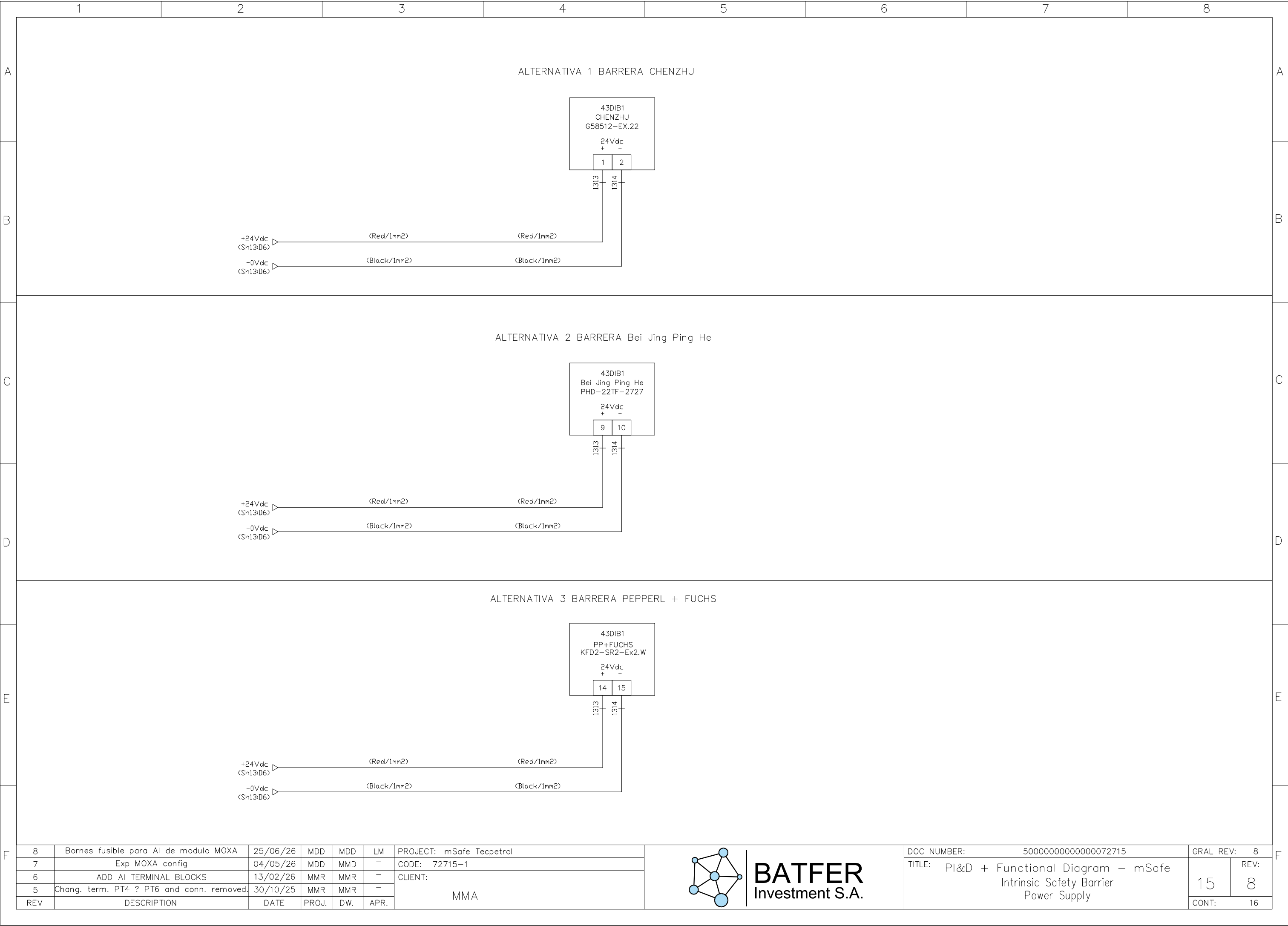


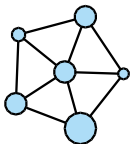
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	TITLE: PI&D + Functional Diagram – mSafe Control Voltage Generation and Distribution		13	REV: 8
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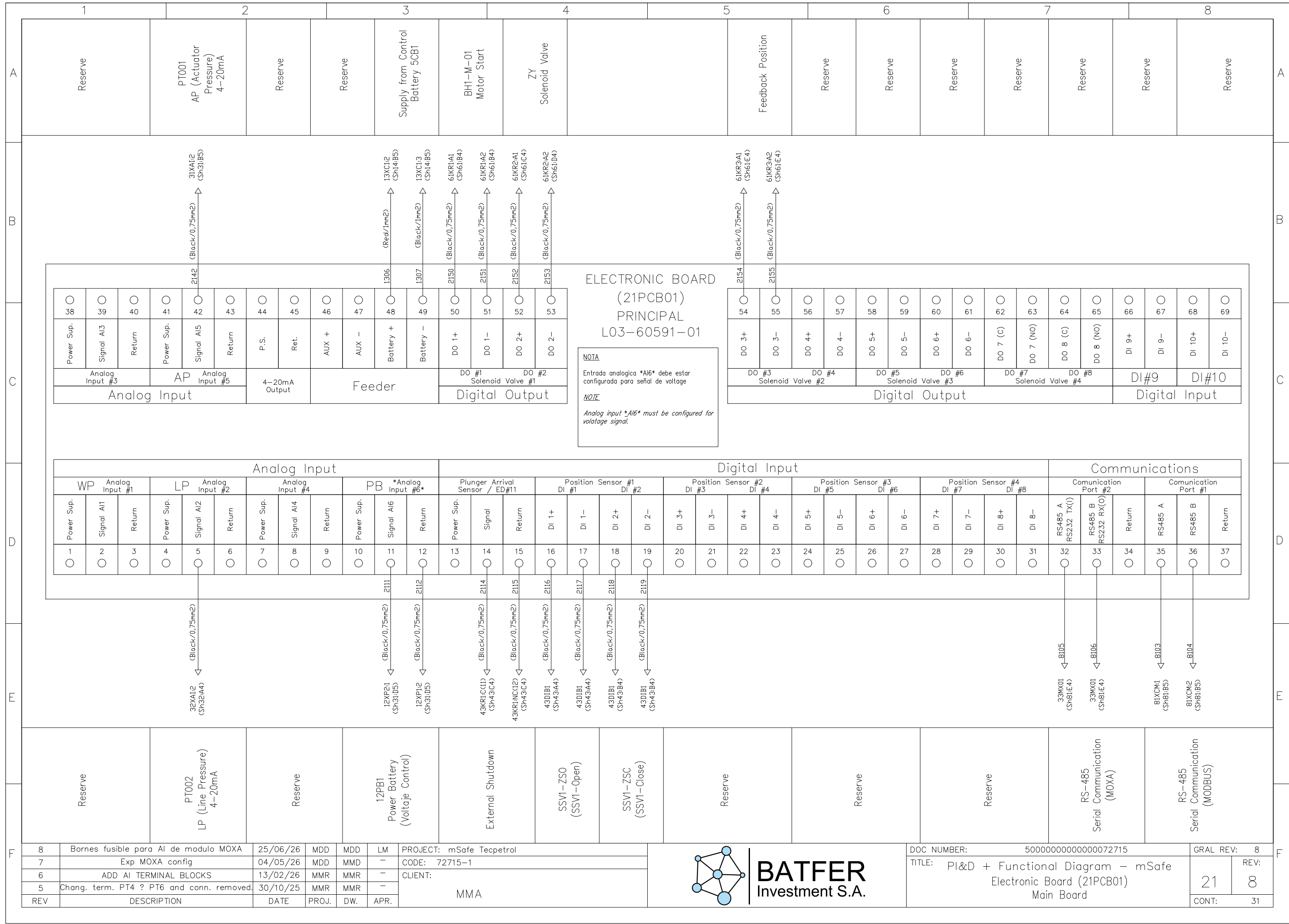


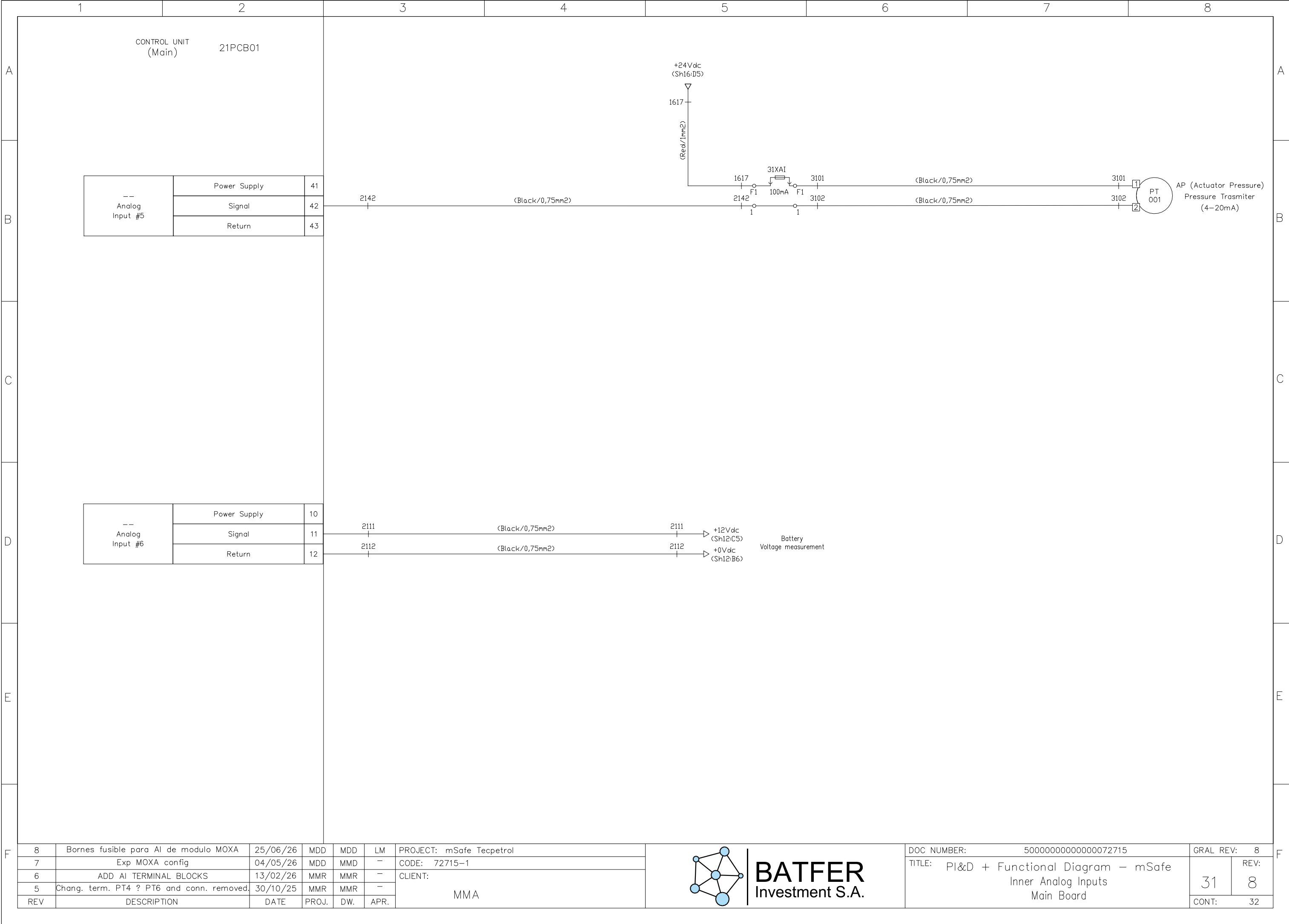
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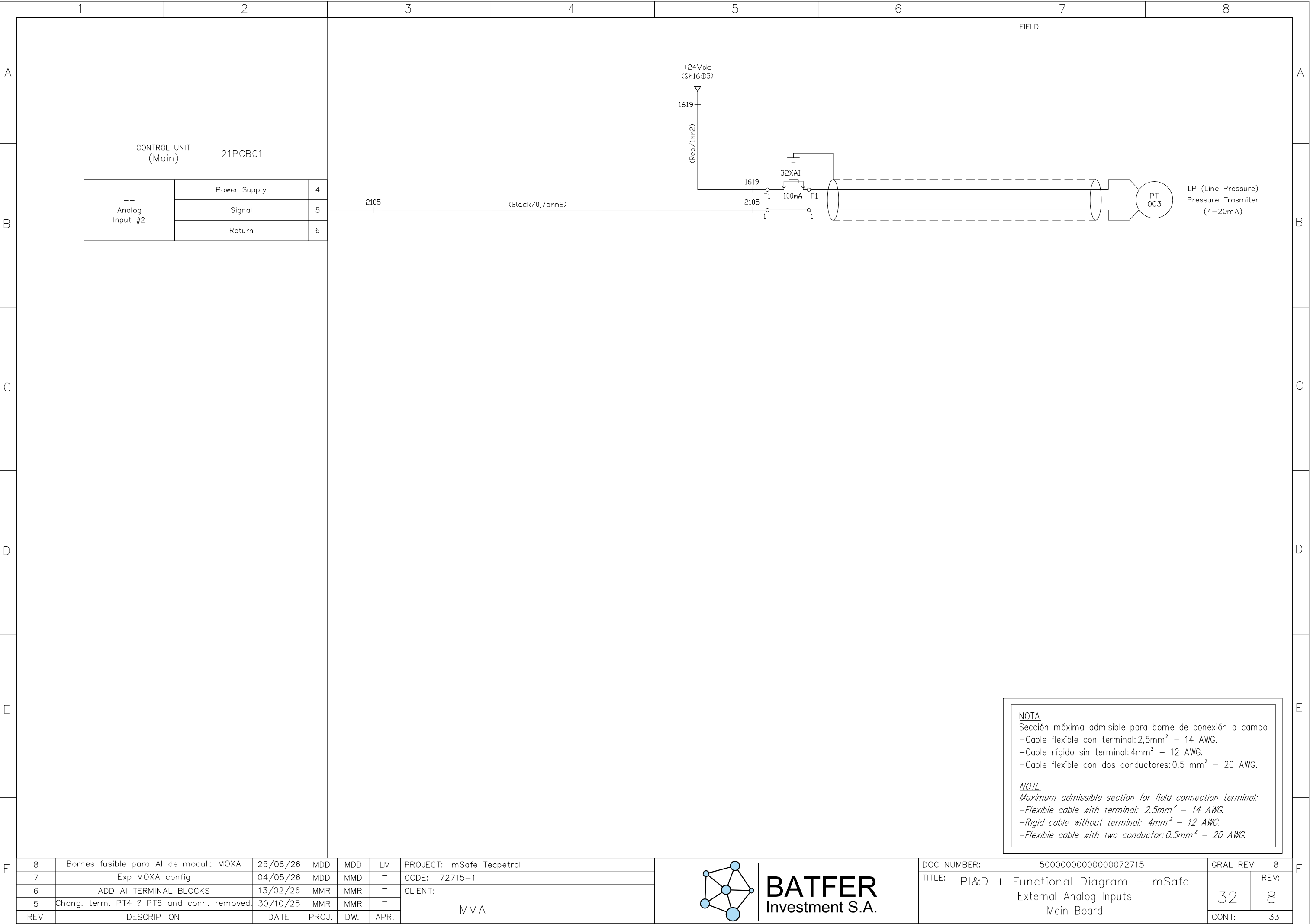
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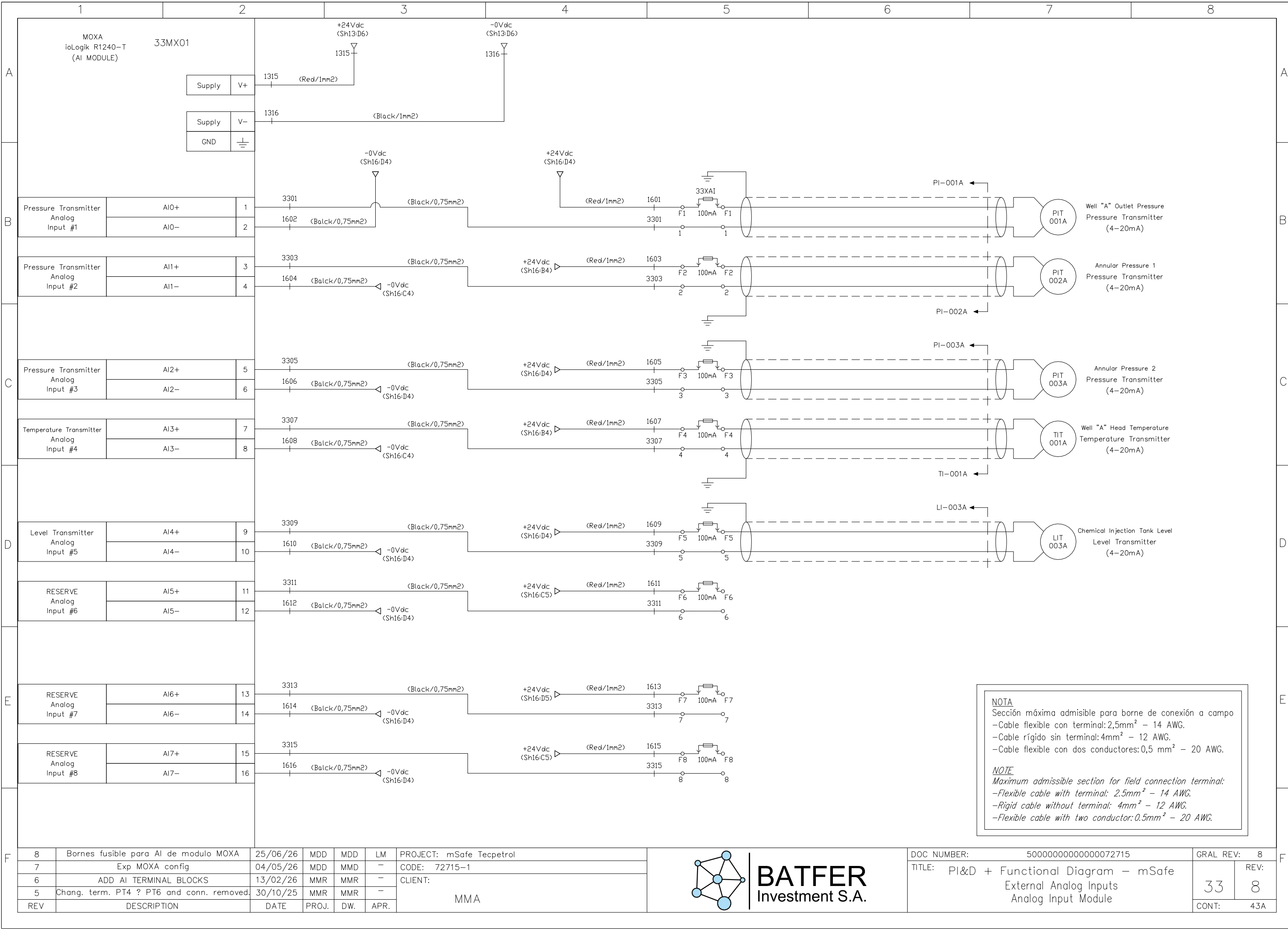




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NOTE: cables not indicated will be black 0.75mm²/18 AWG

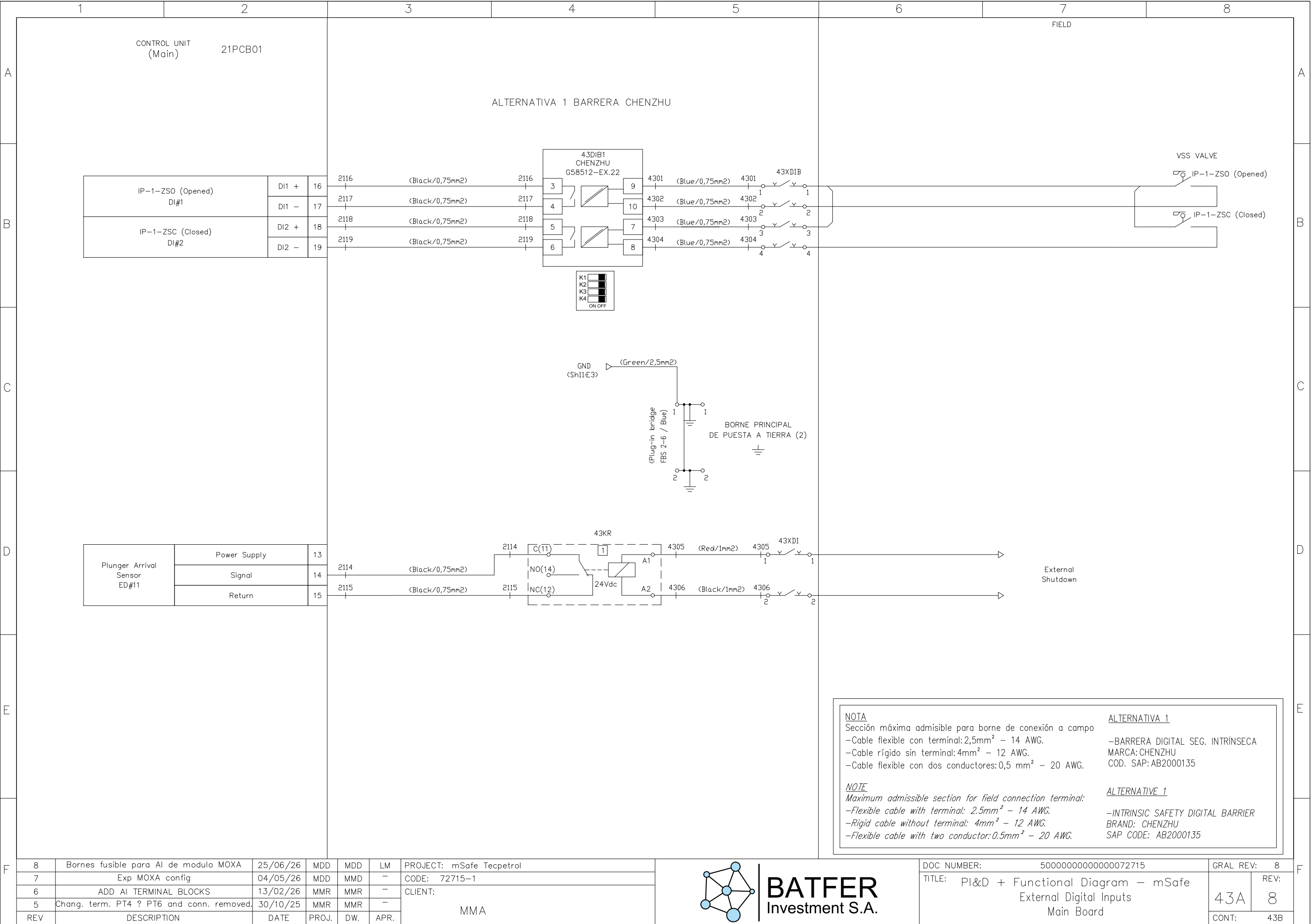


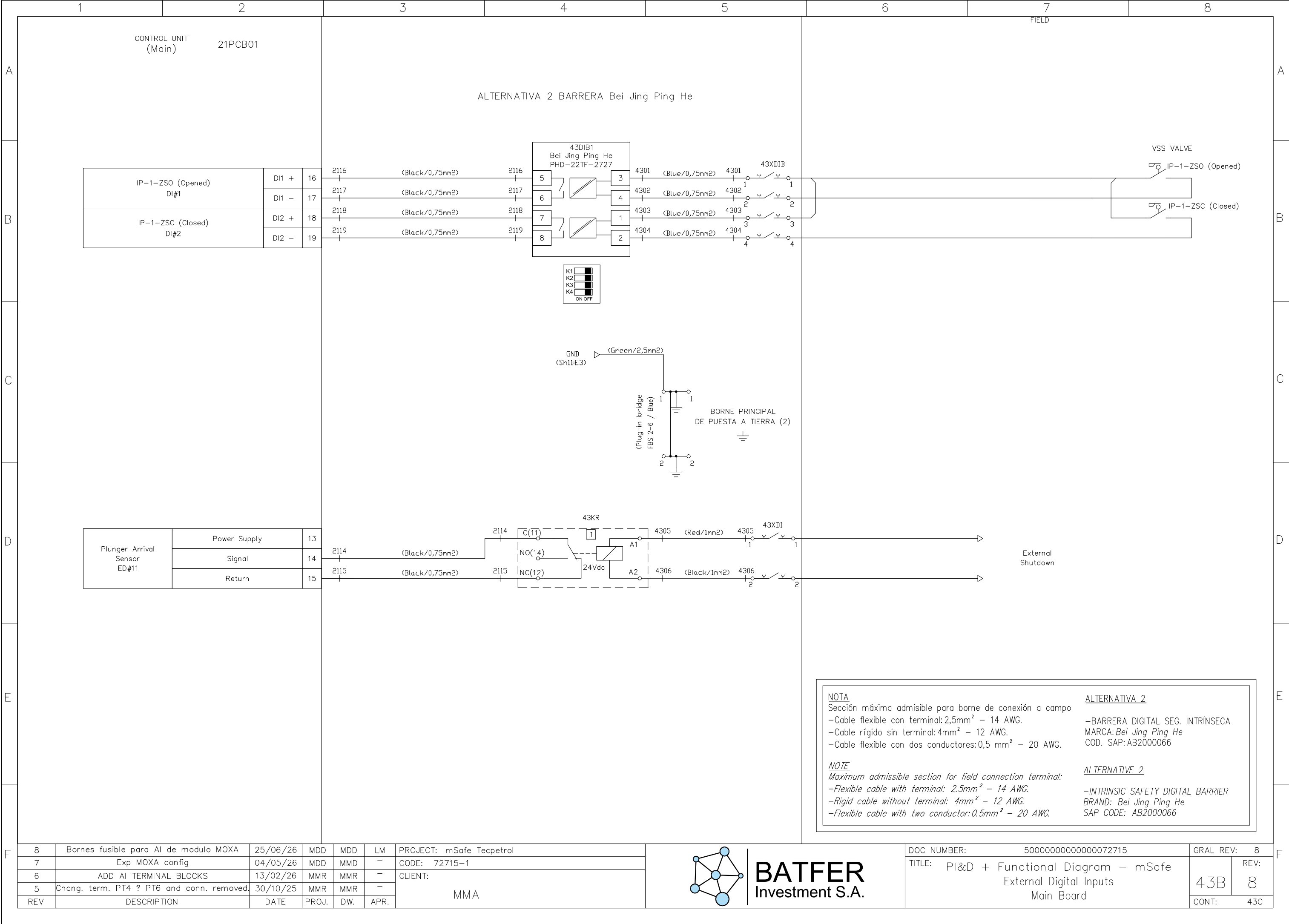


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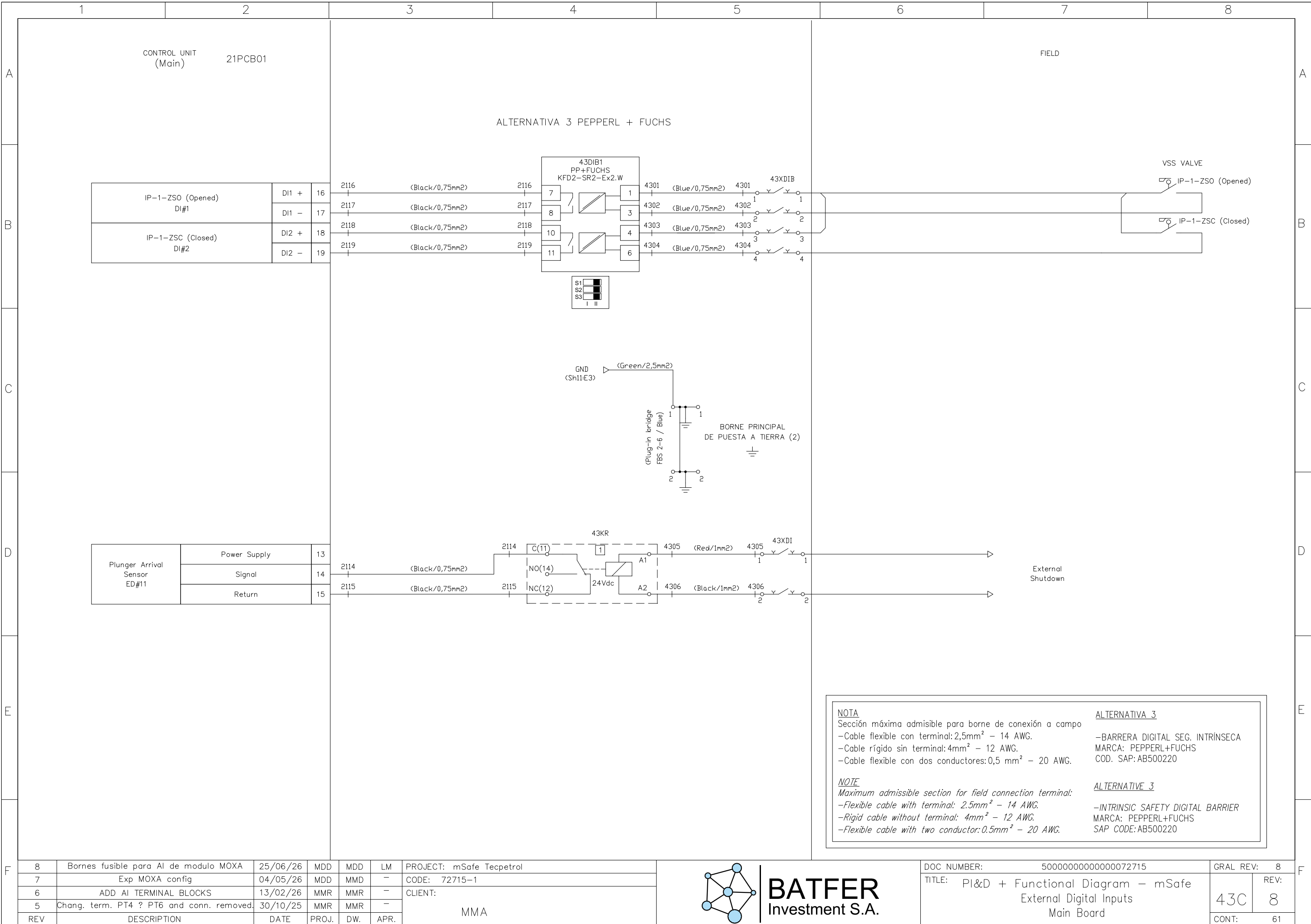
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NOTE: cables not indicated will be black 0.75mm²/18 AWG

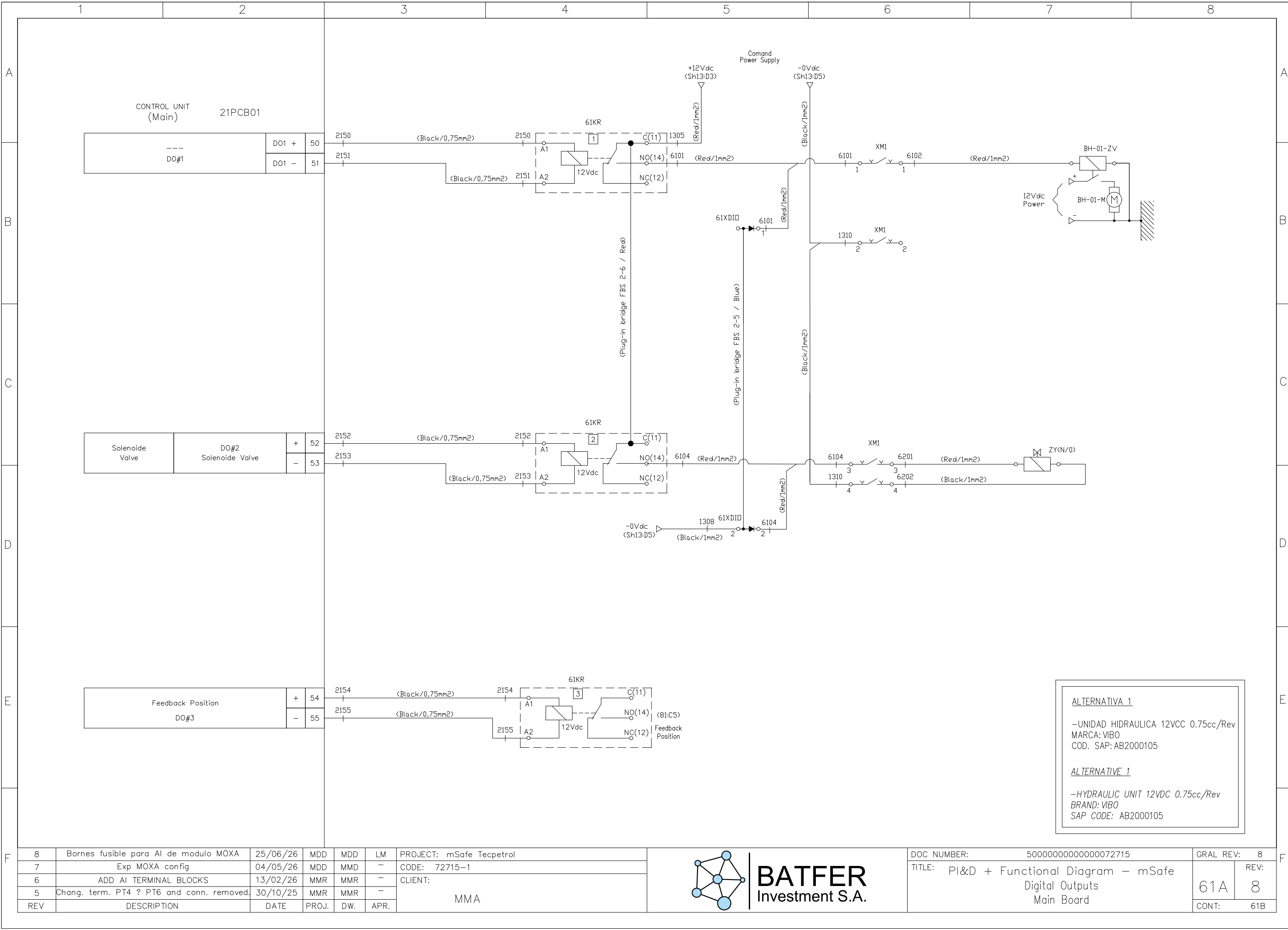




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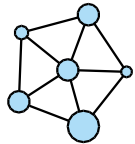
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–UNIDAD HIDRAULICA 12VCC 0.75cc/Rev
MARCA: VIBO
COD. SAP: AB2000105

ALTERNATIVE 1

–HYDRAULIC UNIT 12VDC 0.75cc/Rev
BRAND: VIBO
SAP CODE: AB2000105

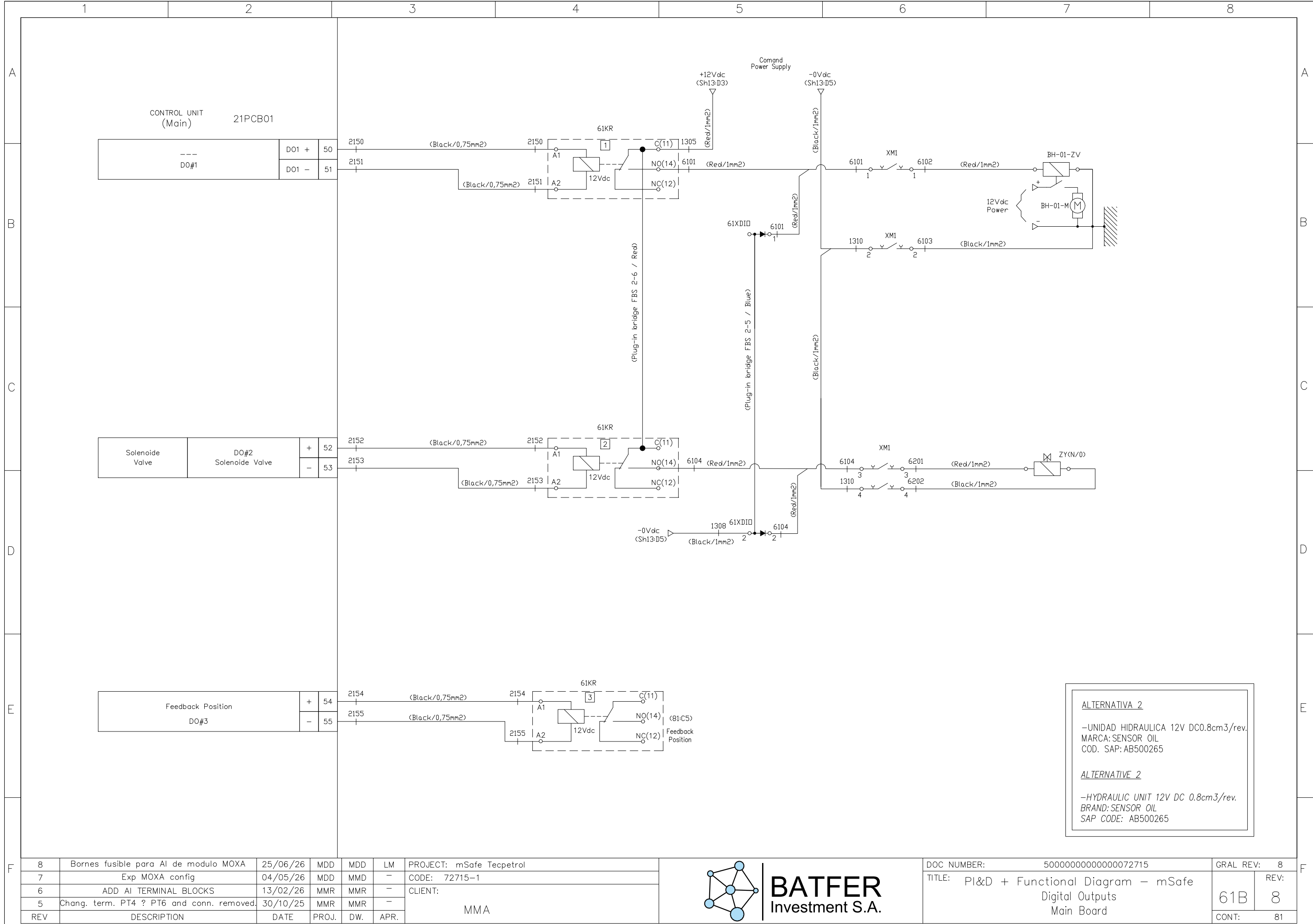
8	Bornes fusible para AI de modulo MOXA	25/06/26	MDD	MDD	LM	PROJECT: mSafe Tecpetrol
7	Exp MOXA config	04/05/26	MDD	MMD	–	CODE: 72715–1
6	ADD AI TERMINAL BLOCKS	13/02/26	MMR	MMR	–	CLIENT:
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REV	DESCRIPTION	DATE	PROJ.	DW.	APR.	MMA

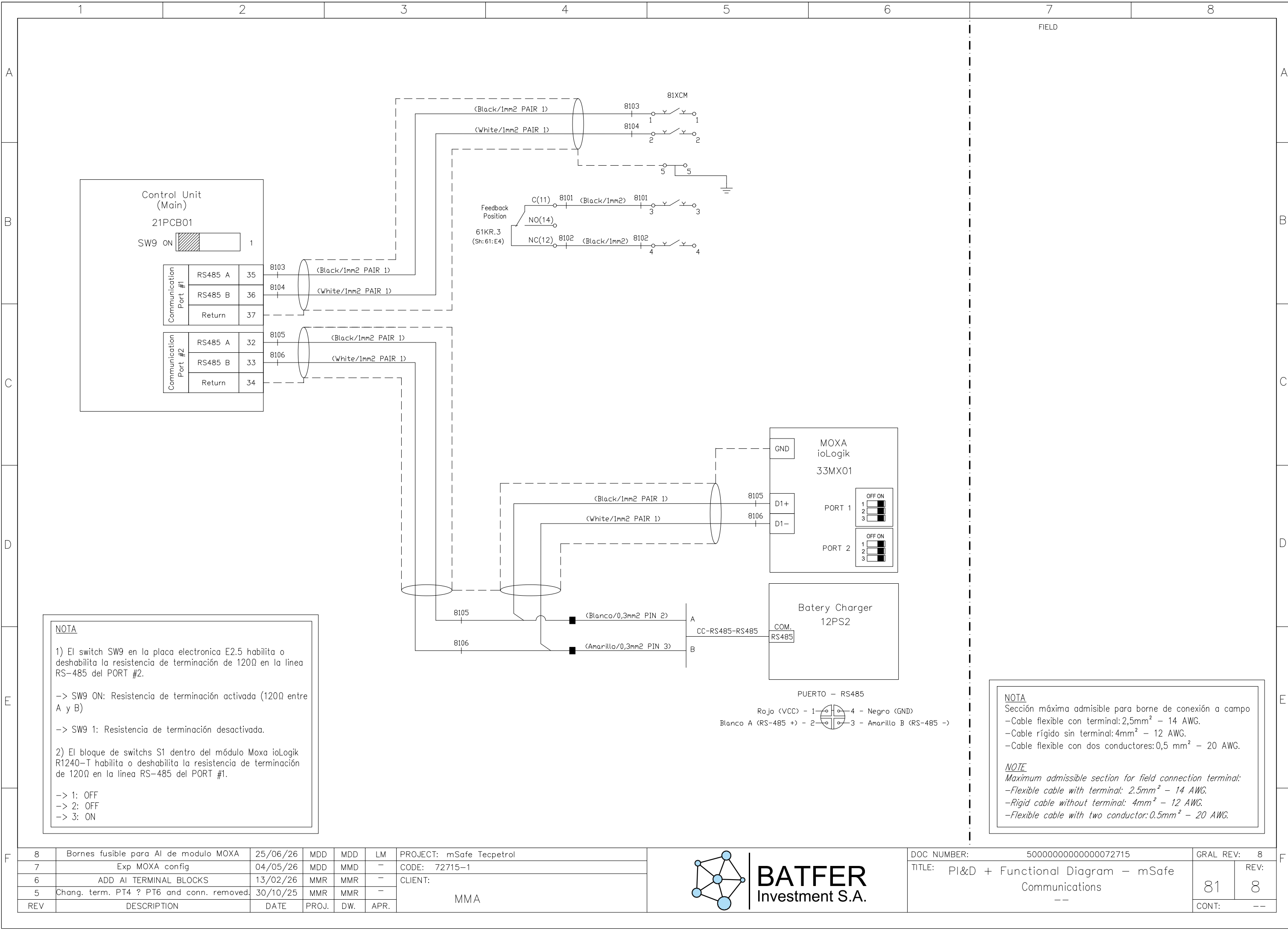


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DOC NUMBER:	5000000000000072715	GRAL REV:	8
TITLE:	PI&D + Functional Diagram – mSafe Digital Outputs Main Board	REV:	8
		CONT:	61B

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